

3/16/2026

Hands-On Email Threat Investigation

EML Analysis and Threat Intelligence
Correlation

Agenda

- 1. Introduction and Incident Response Setup**
- 2. Forensic Analysis and Authentication**
- 3. Attachment and Threat Intelligence**
- 4. Security Framework and Scoring**
- 5. Recommendations, Lessons, and Q&A**

Introduction and Incident Response Setup

Introduction & Project Objective



Project Overview

Forensic analysis of a suspicious .eml file.



Assessment Objective

Evaluate risk and recommend remediation actions.



Incident Summary

Targeted phishing email posed as PayPal invoice.

Incident Response Pipeline

Structured Five-Stage Pipeline

Isolation, header analysis, content extraction, attachment forensics, threat intelligence.

Evidence Integrity & Safety

Analysis performed in a sandboxed environment to prevent contamination.

Reproducible Workflow

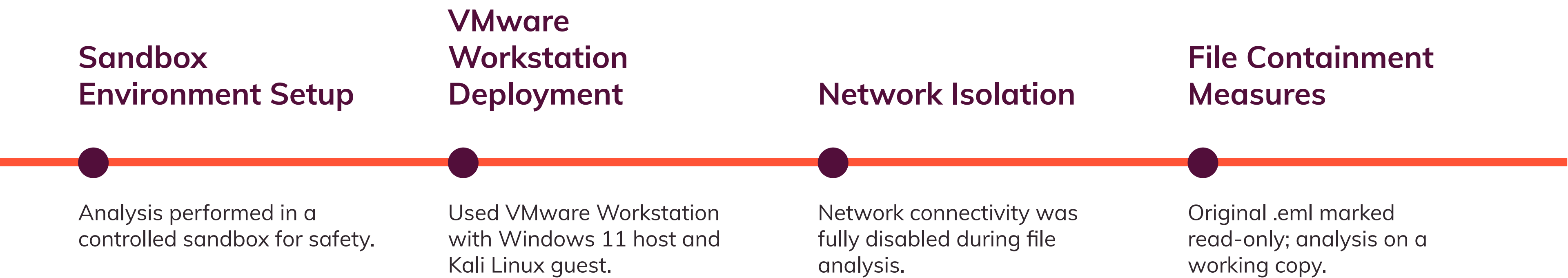
Each stage ensures consistent, repeatable results for forensic investigations.

Comprehensive Threat Assessment

Pipeline covers technical, behavioral, and intelligence-driven analysis.



Environment Isolation & Setup



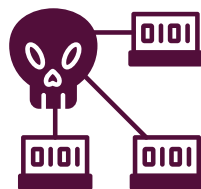
Forensic Analysis and Authentication

Origin & Routing Forensics



Email Origin Traced to Bulgaria

Header analysis revealed dispatch from infected-pc (IP: 45.67.89.101).



Botnet Node Identified

Origin suggests compromised user machine acting as botnet.



Anonymous Swiss Relay Used

Message routed through relay server in Switzerland (IP: 185.203.114.77).



Geographic Routing Visualized

Investigation mapped path: Bulgaria → Switzerland → Final Destination.

Sender Authentication Failures



Spoofed Sender Address

- Threat actor used support@secure-paypal.com to mimic PayPal.



Authentication Protocol Failures

- SPF check failed; sending IP not authorized for domain.
- DKIM signature missing; no cryptographic validation present.
- DMARC policy triggered quarantine due to combined failures.



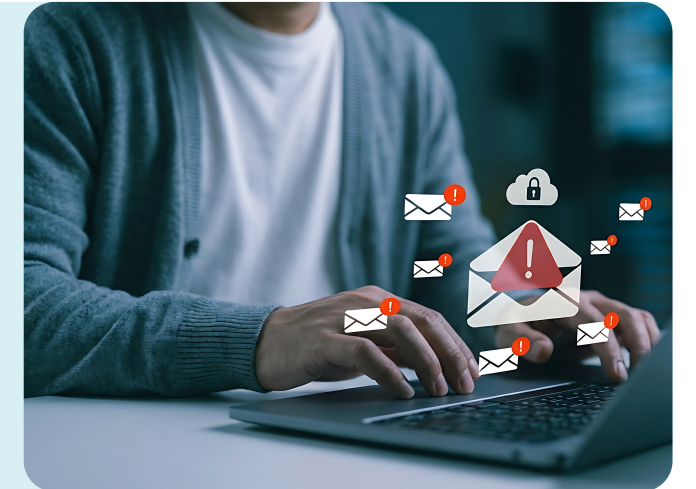
Security Response

- Authentication failures flagged the email for quarantine.

Content & Social Engineering

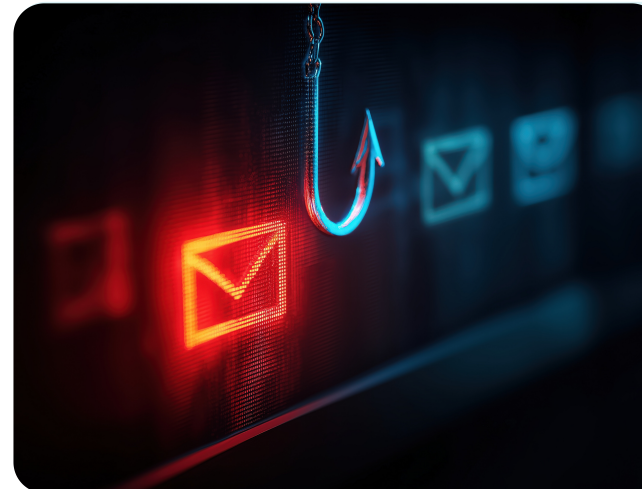
Urgency and Fear Tactics

Subject and body text create pressure to act quickly.



Generic Greeting Signals Mass Attack

Use of 'Dear Customer' reveals non-targeted phishing.



Social Engineering for Immediate Action

Psychological triggers prompt victims to click malicious links.



URL Obfuscation & Extraction

HTML Anchor Tag Deception

Malicious links concealed using anchor tag manipulation.

False Trust via Plain Text

Displayed URL appears legitimate but redirects to phishing site.

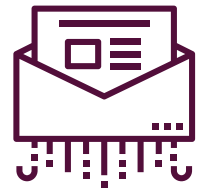
Obfuscated Payload Discovery

HTML analysis exposes hidden redirect to `hxxps[:]//paypalphish[.]vercel[.]app/`



Attachment and Threat Intelligence

Attachment Forensics



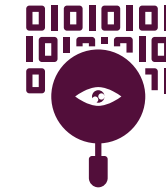
Safe Analysis of Email Attachment

- Strict containment using CLI tools prevented potential threats.



File Type & Payload Discovery

- Attachment was an ASCII text file, not a document.
- Contained only the placeholder string: "You got Hacked".



Cryptographic Evidence Collection

- Generated MD5 and SHA-256 hashes for forensic integrity.

Threat Intelligence Correlation

Phishing URL flagged by VirusTotal

- 17 security vendors marked the URL as **malicious**

Attachment hash analysis

- SHA-256 hash showed **0 detections**, confirming benign file

OSINT infrastructure confirmation

- Domain hosted on Vercel; Abuse.ch confirmed **malicious activity**



Security Framework and Scoring

Security Framework Mapping

Security Framework and Scoring

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MITRE ATT&CK Technique Mapping

- Phishing via obfuscated links (T1566.002) detected.
- Masquerading as PayPal (T1036) to deceive recipients.
- Abuse of cloud infrastructure (T1583.006) for evasion.

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NIST CSF Alignment

- Identify & Detect: Brand impersonation and URL reputation checks.
- Respond & Protect: Network containment and DMARC enforcement.

Automated Threat Scoring



Custom Python-Based Scoring

- Developed a tool to automate threat evaluation using a weighted matrix.



Risk Factor Assessment

- Analyzed attachments, URLs, sender domain, and authentication failures for scoring.



Dashboard Output & Human Override

- System output was **36/100 (Suspicious)**; manual review escalated risk due to cloud abuse.

Scoring Output & Human Correlation



Automated Threat Scoring Result

- System assigned a score of **36/100 (Suspicious)** based on benign attachment.



Manual Risk Assessment

- Human review identified high risk due to abuse of cloud hosting infrastructure.



Human-in-the-Loop Override

- Manual override elevated classification to high-priority incident.

Recommendations, Lessons, and Q&A

Actionable Recommendations



Immediate Threat Containment

Block malicious domains and IPs:
paypalphish.vercel.app,
secure-paypal.com,
45.67.89.101,
185.203.114.77,
216.198.79.3.



Strengthen Email Authentication

Enforce strict DMARC policy (p=reject) to prevent domain spoofing.



Enhance Filtering Controls

Automatically flag emails from newly registered or cloud-hosted domains.



Security Awareness Training

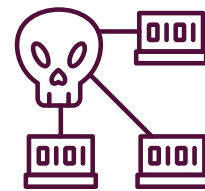
Deploy simulations to teach users to verify link destinations before clicking.

Challenges & Lessons Learned



Safe File Handling Practices

Isolated Linux CLI tools neutralized potential macro threats.



Revealing Hidden Malicious Links

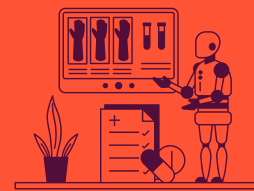
Raw HTML parsing exposed deceptive anchor tag redirection.



Authentication Header Scrutiny

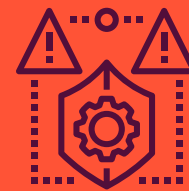
SPF, DKIM, and DMARC analysis dismantled domain spoofing.

Q&A & Key Takeaways



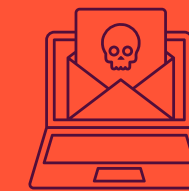
Open Q&A on Investigation Process

- Invite questions about forensic methodology and automation tools.



Layered Analysis & Human Oversight

- Emphasize the value of combining automated scoring with manual review for **accurate threat classification**.



Actionable Steps for Future Response

- Highlight the importance of **network containment**, strict DMARC enforcement, and user awareness training.